James D. Motes

University of Illinois Urbana-Champaign
jmotes2@illinois.edu | jamesmotes.github.io

RESEARCH PROFILE & AGENDA

Robotics and AI researcher developing **adaptive planning systems for human-steerable robot teams**. My work builds robot planning systems that **decompose** large multi-robot problems, **guide** computation to the interactions that require coordination, and **accelerate** the motion validation bottlenecks that limit deployment.

Adaptive Robot-Team Planning: Multi-robot task and motion planning algorithms that coordinate robots only where conflicts, tasks, or workspace structure require it.

Accelerated Planning Infrastructure: Dynamic roadmaps, lazy validation, computational geometry, GPU/SIMD acceleration, and open-source planning software for fast robot motion planning.

Human-Steerable Planning: Planning systems that let people specify, inspect, and refine robot behavior through natural language, AR/VR, and other human-facing interfaces.

ACADEMIC APPOINTMENT

Postdoctoral Researcher, University of Illinois Urbana-Champaign — August 2023 - Present
Advisor: Dr. Nancy M. Amato

EDUCATION

Ph.D. in Computer Science, University of Illinois Urbana-Champaign — August 2023
Ph.D. Thesis: Multi-robot Task and Motion Planning in Hybrid State Spaces
Advisor: Dr. Nancy M. Amato

M.S. in Computer Science, Texas A&M University — August 2019
M.S. Thesis: Interaction Templates for Multi-Robot Systems
Advisor: Dr. Nancy M. Amato

B.S. in Computer Engineering, Texas A&M University — May 2018
Minor in Mathematics, Engineering Honors, Undergraduate Research Scholar

RESEARCH LEADERSHIP HIGHLIGHTS

- Lead several graduate research teams in the Parasol Lab across multi-robot planning, accelerated motion validation, human-steerable planning, and open-source planning infrastructure.
- Set project directions, coordinate research milestones, mentor paper development, and integrate related projects into a coherent agenda around adaptive planning for robot teams
- Mentor over 20 graduate students, including six now graduated and five now PhD candidates; mentoring has contributed to 14 published manuscripts and another 6 now in review.
- Primary author on NSF and industry collaboration research proposals
- Coordinate collaborations spanning robotics, NLP, HRI, computational geometry, industrial automation, and planning systems

SELECTED TECHNICAL CONTRIBUTIONS

Decomposable Multi-Robot Planning: Developed and extended hypergraph-based planning frameworks for multi-robot task and motion planning, including DaSH[P5], Lazy-DaSH[P14], and guidance informed hypergraph planning[P12].

Adaptive Robot Coordination: Developed multi-robot motion planning strategies that dynamically couple and decoupled robots around local coordination conflicts[P7], with extensions to kinodynamic planning[P10], experience reuse[U6], parallelization[U2], and asymptotic optimality[U3].

Accelerated Motion Validation: Developed methods for reducing motion validation bottlenecks using computational geometry[P18], lazy validation[P14], multi-core CPU parallelization[U2], and GPU/SIMD serialization[U2,U5] for dynamic and multi-robot environments.

Human-Steerable Planning Interfaces: Developed planning tools that expose planning controls to non-expert users through interfaces like natural language[IP2] and AR/VR systems[U4].

PUBLICATIONS & SCHOLARLY OUTPUT

Publication Overview

- 14 peer reviewed archival publications, including 11 journal articles and 3 refereed conference proceedings.
- 3 first author peer reviewed publications and 3 first author manuscripts under review.
- 10 student-led publications directly mentored with another 3 under review and 4 in preparation.
- Additional outputs include 11 preprints, technical reports, workshop papers, and extended abstracts.
- Google Scholar: 376 citations, h-index: 7, as of 06/19/2026

Notation.

- Name is bolded: **James D. Motes**
- * indicates equal contribution
- † indicates a student or junior researcher directly mentored: A. Student†
- ‡ indicates a student-led paper for which I served as primary day-to-day research mentor or project lead: **James D. Motes**‡

Research Theme Map

- Adaptive Robot-Team Planning: [P1-5], [P7], [P10], [P12-14], [U3], [U6], [IP3-4]
- Accelerated Planning Infrastructure: [P4], [P8], [U1-2], [U5]
- Human-Steerable Planning: [U1], [IP1-2]

Peer-Reviewed Archival Publications

Journal Articles

[P14] S. Lee†, **James D. Motes**‡, I. Ngui†, M. Morales, and N. M. Amato, “Lazy-DaSH: Lazy Approach for Hypergraph-based Multi-Robot Task and Motion Planning,” *IEEE Transactions on Robotics (T-RO)*, to appear, 2026.

[P12] C. McBeth†, **James D. Motes**‡, I. Ngui†, M. Morales, and N. M. Amato, “Scalable Multi-Robot Motion Planning Using Guidance-Informed Hypergraphs,” *IEEE Robotics and Automation Letters (RA-L)*, 2026.

[P11] H. Lee†, **James D. Motes**‡, M. Morales, and N. M. Amato, “An Analysis of Constraint-Based Multi-Agent Pathfinding Algorithms,” *IEEE Transactions on Robotics (T-RO)*, 2025. Presented at the *IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, June 2026.

[P10] M. Qin†, I. Solis†, **James D. Motes**‡, M. Morales, and N. M. Amato, “K-ARC: Adaptive Robot Coordination for Multi-Robot Kinodynamic Planning,” *IEEE Robotics and Automation Letters (RA-L)*, 2025. Presented at the *IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, June 2026.

[P7] I. Solis†, **James D. Motes**‡, M. Qin†, M. Morales, and N. M. Amato, “Adaptive Robot Coordination: A Subproblem-based Approach for Hybrid Multi-Robot Motion Planning,” *IEEE Robotics and Automation Letters (RA-L)*, 2024. Presented at *IEEE ICRA@40*, Rotterdam, The Netherlands, Sept. 2024.

[P6] C. McBeth†, **James D. Motes**‡, D. Uwacu, M. Morales, and N. M. Amato, “Scalable Multi-Robot Motion Planning for Congested Environments With Topological Guidance,” *IEEE Robotics and Automation Letters (RA-L)*, 2023. Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, May, 2024.

[P5] **James D. Motes**, T. Chen, T. Bretl, M. M. Aguirre, and N. M. Amato, “Hypergraph-Based Multi-Robot Task and Motion Planning,” *IEEE Transactions on Robotics (T-RO)*, 2023. Presented at the *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, May 2024.

[P4] H. Lee†, **James D. Motes**‡, M. Morales, and N. M. Amato, “Parallel Hierarchical Composition Conflict-Based Search for Optimal Multi-Agent Pathfinding,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 6, no. 4, pp. 7001–7008, 2021. Presented at *IEEE International Conference on Intelligent Robots and Systems (IROS)*, Prague, Czech Republic, 2021 (virtual).

[P3] I. Solis†, **James D. Motes**‡, R. Sandström, and N. M. Amato, “Roadmap-Optimal Multi-Robot Motion Planning Using Conflict-Based Search,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 6, no. 3, pp. 4608–4615, 2021. Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Xi’an, China, 2021 (virtual).

[P2] **James D. Motes**, R. Sandström, H. Lee†, S. Thomas, and N. M. Amato, “Multi-Robot Task and Motion Planning With Subtask Dependencies,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 5, no. 2, pp. 3338–3345, 2020. Presented at *IEEE International Conference on Robotics and Automation, Paris (ICRA)*, France, 2020 (virtual).

[P1] **James D. Motes**, R. Sandström, W. Adams, T. Ogunyale†, S. Thomas, and N. M. Amato, “Interaction Templates for Multi-Robot Systems,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 4, no. 3, pp. 2926–2933, 2019. Presented at *IEEE International Conference on Intelligent Robots and Systems (IROS)*, Macao, China, 2019.

Refereed Conference Proceedings

[P13] I. Ngui†, C. McBeth†, **James D. Motes**‡, M. Morales, and N. M. Amato, “Scalable Multi-Robot Motion Planning via Hierarchical Subproblem Expansion and Workspace Decomposition Refinement,” in *Algorithmic Foundations of Robotics XVII: Proc. the Seventeenth World Symposium on the Algorithmic Foundations of Robotics (WAFR)*, 2026.

[P9] A. Goddu, T. Brown, M. Paton, **James D. Motes**, and T. Chen, “Modifying ABIT* for Tethered Rappelling Robot Motion Planning,” in *Proc. 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, pp. 722–728, 2025.

[P8] S. Ashur†, M. Lusardi, M. Markowicz†, **James D. Motes**‡, M. Morales, S. Har-Peled, and N. M. Amato, “SPITE: Simple Polyhedral Intersection Techniques for Modified Environments,” in *Algorithmic Foundations of Robotics XVI: Proc. the Sixteenth Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2024.

Manuscripts Under Review

First Author

[U3] **James D. Motes**, M. Morales, N. M. Amato, “AO-ARC: Almost-Surely Asymptotically Optimal Multi-Robot Motion Planning with ARC.”

[U2] **James D. Motes**, M. Morales, N. M. Amato, “P-ARC: Exploiting Subproblem Independence for Parallel Multi-Robot Motion Planning.”

[U1] **James D. Motes**, M. Morales, N. M. Amato, “Multi-Robot Motions in Milliseconds: Vector-Accelerated Primitives for Sampling-Based Planning,” *arXiv* preprint arXiv:2604.23960, 2026.

Mentored Student-Led Manuscripts

[U6] I. Solis†, **James D. Motes**‡, J. Montalvo, M. Qin†, M. Morales, and N. M. Amato, “Experience-Based Subproblem Planning for Multi-Robot Motion Planning,” *arXiv* preprint arXiv:2411.08851, 2024.

[U5] Y. Arad†, M. Markowicz†, S. Ashur†, **James D. Motes**‡, M. Morales, and N. M. Amato, “Serialized Red-Green-Gray: A Quick Heuristic Validation of Edges in Dynamic Roadmap Graphs.”

[U4] I. Ngui†, C. McBeth†, A. Santos, G. He, K. J. Mimnaugh, **James D. Motes**, L. Soares, M. Morales, and N. M. Amato, “ERUPT: An Open Toolkit for Interfacing with Robot Motion Planners in Extended Reality,” *arXiv* preprint arXiv:2510.02464, 2025.

Manuscripts in Preparation

[IP4] S Lee†, **James D. Motes**‡, M. Morales, and N. M. Amato, “PDDL-DaSH: Expanding Hypergraph Planning With Explicit and Implicit Query Strategies.

[IP3] M. Markowicz†, **James D. Motes**‡, I. Solis†, M. Morales, and N. M. Amato, “MR-SPITE: Leveraging Dynamic Roadmap Graphs in Multi-Robot Motion Planning.”

[IP2] A. Yammanuru†, J. Buabeng†, J. Liu†, A. Luo†, **James D. Motes**‡, Marco Morales, and Nancy M. Amato, “Object-centric Reasoning for Behavior with Instruction-conditioned Trajectories.”

[IP1] I. B. Love†, D. J. Lohan, **James D. Motes**, E. M. Dede, M. Morales, and N. M. Amato, “A Hierarchical Approach for the Generation of Varied Cable Harness Designs.”

Additional Scholarly Output

Preprints and Technical Reports

[TR2] H. Lee†, Z. Serlin, **James D. Motes**, B. Long, M. Morales, and N. M. Amato, “PRISM: Complete Online Decentralized Multi-Agent Pathfinding With Rapid Information Sharing Using Motion Constraints,” *arXiv* preprint arXiv:2505.08025, 2025.

[TR1] A. Attali, S. Ashur†, I. B. Love†, C. McBeth†, **James D. Motes**, D. Uwacu, M. Morales, and N. M. Amato, “A Framework for Guided Motion Planning,” *arXiv* preprint arXiv:2404.03133, 2024.

Workshop Papers and Extended Abstracts

[W9] M. Markowicz†, S. Ashur†, **James D. Motes**‡, and N. M. Amato, “Semi-Lazy Rearrangement Solver for Accelerating Task Planning Using SPITE Dynamic Roadmaps,” in *Extended Abstract, 4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments (ICRA 2025)*, Atlanta, GA, USA, May 19, 2025.

[W8] H. Lee†, **James D. Motes**‡, Z. Serlin, M. Morales, and N. M. Amato, “Distributed Constraint-Based Search Using Multi-Hop Communication,” in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W7] I. Ngui†, **James D. Motes**‡, H. Lee†, M. Morales, and N. M. Amato, “Ergonomic Motion Planning for Human-Robot Collaboration,” in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W6] K. Elimelech,* **James D. Motes,*** M. Morales, N. M. Amato, M. Vardi, and L. E. Kavraki, “Encoding Reusable Multi-Robot Planning Strategies as Abstract Hypergraphs,” in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W5] S. Lee†, **James D. Motes**‡, I. Ngui†, M. Morales, and N. M. Amato, “Lazy-DaSH: Lazy Approach for Hypergraph-based Multi-Robot Task and Motion Planning,” in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W4] M. Lusardi,* M. Markowicz†,* S. Ashur†, **James D. Motes**, M. Morales, S. Har-Peled, and N. M. Amato, "SPITE: Simple Polyhedral Intersection Techniques for Modified Environments," in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W3] I. Ngui†, S. Lee†, **James D. Motes**‡, M. Morales, and N. M. Amato, "A Hierarchical Approach to Workstation-based Task-Allocation and Motion Planning," in *Extended Abstract, RAFF2023: Robotics & AI in Future Factory (IROS 2023 Workshop), IROS 2023*, Detroit, MI, USA, Oct. 1–5, 2023.

[W2] A. Attali, S. Ashur†, I. B. Love†, C. McBeth†, **James D. Motes**, D. Uwacu, M. Morales, and N. M. Amato, "Evaluating Guiding Spaces for Motion Planning," in *Extended Abstract, Evaluating Motion Planning Performance, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022)*, Kyoto, Japan, Oct. 23, 2022.

[W1] T. Chen, Z. Huang, **James D. Motes**, J. Geng, Q. Ta, H. Dinkel, H. Abdul-Rashid, J. Myers, Y. Mun, W. Lin, Y. Huang, S. Liu, M. Morales, N. M. Amato, K. Driggs-Campbell, and T. Bretl, "Insights from an Industrial Collaborative Assembly Project: Lessons in Research and Collaboration," in *IEEE ICRA Workshop on Collaborative Robots and the Work of the Future*, Philadelphia, USA, 2022.

GRANT & SPONSORED RESEARCH DEVELOPMENT

National Science Foundation (NSF) - Foundational Research in Robotics (FRR) Proposal September 2025

- Primary author on proposal to customize robotic motion behaviors for non-expert users through natural language preference specification
- Coordinated collaboration with natural language processing researchers at the University of Illinois Urbana-Champaign
- Developed research narrative connecting human preference specification, motion planning, and human-steerable robot behavior.

National Science Foundation (NSF) - Engineering Research Center (ERC) Site Visit March 2026

- Supported preparation for ERC site visit with external reviewers
- Developed research narrative and presentation for research thrust

Industry Proposals and Whitepapers April 2025 - Present

- Primary author on eight proposals and secondary author on four more
- Proposal topics include human-robot interaction, multi-robot planning, environment design, and data-driven planning
- Developed sponsored research narratives connecting adaptive robot team planning to deployable industrial automation systems

INVITED TALKS & SEMINARS

Lessons from Manufacturing Startups

- AI Meets Robots & Machines Seminar, University of Illinois Urbana-Champaign, Urbana, Il.
- February 2026. Invited seminar.

Multi-Robot Task and Motion Planning in Hybrid State Spaces

- Kavraki Lab, Department of Computer Science, Rice University, Houston, TX
- February 2024. Invited Lab Seminar

Planning Motions and Tasks for Manipulators, Multi-Robot Systems and Biomolecules

- Center for Autonomy Portfolio Discovery Workshop, University of Illinois Urbana-Champaign, Urbana, IL
- October 2019. Invited Workshop Talk.

TEACHING & CURRICULUM DEVELOPMENT

Lead Instructor, AI4ALL August 2021 - December 2023

- Instructor for introductory artificial intelligence courses delivered through partner programs at the University of Texas at El Paso, New Mexico State University, Worcester Polytechnic Institute, and Texas A&M University.
- Developed curriculum for the Discover AI introductory artificial intelligence course.
- Led and mentored instructor teams across participating universities.
- Teaching assistant for Discover AI at the University of Illinois Urbana-Champaign in Fall 2021.

Teaching Assistant, University of Illinois Urbana-Champaign January 2021 - May 2021

- Teaching assistant for Computer Science Ph.D. seminar course.
- Organized guest speakers, managed grading, and coordinated student assignments.

Internal Lab Instruction, Parasol Lab August 2019 - May 2021

- Developed and taught a motion-planning course for new Parasol Lab members.

MENTORING, ADVISING & OUTREACH

Postdoctoral Researcher, Parasol Lab, University of Illinois Urbana-Champaign August 2023 - Present

- Mentor over 20 graduate students across multi-robot planning, guided planning, accelerated validation, and human-steerable planning projects.
- Supported 6 graduate student completions and 5 current PhD candidates.
- Postdoctoral mentoring has contributed to 7 student-led publications and another 4 manuscripts under review.
- Coordinate student research milestones, paper development, software integration, and cross-project planning infrastructure.

Graduate Researcher, University of Illinois Urbana–Champaign, Texas A&M University August 2018 - August 2023

- Mentored 14 graduate students within the Parasol Lab research group (many continued in postdoc mentoring).
- Mentored undergraduate researchers, including local university students and DREU program participants.
- Graduate student mentoring contributed to 4 student-led publications.
- Taught motion planning to new lab members through an internal Parasol Lab course.

Houston Robotics Club August 2023 - Present

- Mentor community college and high school students on AI and robotics projects.
- Supported students who later enrolled in STEM undergraduate programs or obtained internships with local robotics companies.

PROFESSIONAL SERVICE

Program Committee Member

- World Symposium on Algorithmic Foundations of Robotics, 2026.
- World Symposium on Algorithmic Foundations of Robotics, 2024.

Reviewer/Referee August 2018 - Present

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Robotics (T-RO)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Symposium on Multi-robot and Multi-agent Systems (MRS)
- World Symposium (formerly International Workshop) on Algorithmic Foundations of Robotics (WAFR)
- International Journal of Robotics Research (IJRR)
- Artificial Intelligence Journal (AIJ)

Professional Memberships

- IEEE Robotics and Automation Society May 2019 - Present

INDUSTRY, ENTREPRENEURSHIP & RESEARCH TRANSLATION

Industrial Human-Robot Factory Collaboration, UIUC August 2019 - August 2023

- Collaborated with Foxconn Interconnect Technologies and UIUC research groups led by Timothy Bretl and Katherine Driggs-Campbell on collaborative human-robot factory projects.
- Connected multi-robot task and motion planning research to industrial assembly and human-robot collaboration settings.

Normandy Automation, Founder May 2023 – August 2025

- Developed autonomous robotic manufacturing concepts for just-in-time manufacturing with minimal human reprogramming.
- Leveraged AI and multi-robot systems for autonomous welding and CNC machine-tending workflows.
- Wrote NSF SBIR proposals for autonomous robotic manufacturing systems.

Optigon Inc., Principal Software Engineer May 2024 – Present

- Lead software development for DOE SBIR-funded high-throughput, non-contact spectroscopic metrology and analysis tools for clean-energy materials, including perovskite photovoltaics.
- Develop machine learning analytical models of solar cell behavior.

HONORS

Engineering Graduate Merit Fellowship	August 2018 - August 2019
Presidential Endowed Scholarship	August 2014 - May 2018
Distinguished Student Award - Dwight Look College of Engineering	May 2017
Industrial Affiliates Program Scholarship	August 2014 - May 2018
Dell Merit Scholarship	August 2017 - May 2018
Salutatorian, Rowlett High School (2/591)	June 2014